University of Pretoria
Department of Mathematics and Applied Mathematics

WTW 158 Calculus

ASSIGNMENT: SEMESTER TEST 2

5 June 2021

When submitting this assignment, you agree to the following:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

TIME: 90 minutes for the ClickUP test and assignment.
MARKS: 22 for this assignment

INSTRUCTIONS:

1. The assignment consists of 1 page and 5 questions.

2. Draw a line with a ruler at the end of each question.

3. Write your surname, initials and student number clearly in the beginning of your assignment.

4. **Show all your steps and calculations**.

5. You have 90 minutes to do the whole test: ClickUP test and assignment.

6. You have 30 minutes for downloading the assignment, to prepare the assignment for submittance and to submit.

7. Only handwritten assignments submitted in one PDF file will be accepted.

8. Before submitting, make sure that you are satisfied with your answers and that you submit the correct document.

9. No late assignments will be accepted.

10. No assignments sent via e-mail will be accepted.

11. Calculators **may not** be used.

Question 1 [2,1,1]
Consider the function
$$f(x) = \frac{\ln(x) + \sqrt{x-2}}{x-3}.$$

(1) Determine the interval on which f is continuous. Explain your answer.

(2) Determine the limit (if it exists)
$$\lim_{x \to 11} f(x).$$

(3) Determine the limit (if it exists)
$$\lim_{x \to 3^+} f(x).$$

Question 2 [2,3]
Clearly show how you use the limit laws and write down all your steps.

(1) Evaluate the limit (if it exists):
$$\lim_{t \to 0} \frac{\tan(6t)}{\sin(2t)}.$$

(2) Evaluate the limit (if it exists):
$$\lim_{x \to -\infty} \left(\sqrt{x^2 - 2x + 4} + x \right)$$

Question 3 [4 marks]
Find the domain and all the asymptotes of the graph of the function
$$f(x) = \frac{x^2 - 1}{x(x+1)}.$$
Show all your calculations and use interval notation.

Question 4 [2,2,3]
Clearly write down all your steps to show how you apply the necessary rules and laws.

(1) Let $a > 0$. Find the derivative of
$$f(x) = \tan^{-1}(3x) + a^{\sqrt{x}}.$$
Do not simplify your answer.

(2) Consider
$$f(x) = \frac{\ln(x+1)}{x^4}.$$
Determine f' and the domain of f'. Use interval notation.

(3) Find the derivative of
$$f(x) = (\sin(x))^{\cosh(x)}.$$

Question 5 [2 marks]
By the *end behavior* of a function we mean the behavior of its values as $x \to \infty$ and as $x \to -\infty$. Two functions have the *same end behavior* if their ratio approaches 1 as $x \to \infty$. Show that P and Q have the same end behavior if
$$P(x) = 6x^4 - 5x^3 - 2x \quad \text{and} \quad Q(x) = 6x^4.$$